→ union, Intersection, Difference.
($\cup$) ($\cap$) ($-$)

$2+2$ $3-1$ $2 \cdot 4$ $6/5$

operations

→ cartesian product - $A \times B = \{(a,b) : \dots\}$

for any two set A and B :-

$A \cup B = \{x : x \in A \text{ or } x \in B\}$

$A \cap B = \{x : x \in A \text{ and } x \in B\}$

$A - B = \{x : x \in A \text{ and } x \notin B\}$

difference

ex:- $A = \{a, b, c, d, e\}$ $B = \{d, e, f\}$ $C = \{1, 2, 3\}$

$A \cup B = \{a, b, c, d, e, f\}$

$A \cap B = \{d, e\}$

$A - B = \{a, b, c\}$

$B - A = \{f\}$

$A \cup B = B \cup A$
$A \cap B = B \cap A$
$A - B \neq B - A$

→ $(A \cap B) \times B = ?$ $\{d, e\} \times \{d, e, f\}$
$\{(d,d), \dots\}$

→ $(A \cap C) \times B = ?$

$\emptyset \times B \Rightarrow \emptyset$